

Bicommutant Categories from Conformal Nets

Towards constructing fully extended functorial (Segal) chiral CFTs

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Chiral Conformal Field Theory

Physics: A chiral CFT is a structure that captures the data of the chiral part of a full CFT, i.e., it captures the assignment of the vector space of conformal blocks to a Riemann surface with genus g with n marked points that assembles into a holomorphic bundle over $\overline{\mathcal{M}}_{g,n}$.

Axiomatic approaches to Chiral CFT

Analytic/Algebraic chiral CFT

Segal χ -CFT

Vertex Operator Algebras (VOA)

Conformal Nets

It is conjectured that both unitary VOAs and Conformal Nets capture equivalent data for a chiral CFT, proven under some extra assumptions [CKLW18, HT22].

Our aim to give a *construction of a functorial 2d χ -CFT* from the algebraic data of a Conformal Net.

Conformal Nets

A conformal net is defined by the *data* of a

- Hilbert space $\mathcal{H}_0$ called *the vacuum sector*.
- a continuous projective unitary representation u of $\text{Diff}^+(S^1)$ on $\mathcal{H}_0$ which restricts to an honest representation on the Möbius subgroup.
- vector $\Omega \in \mathcal{H}_0$ called *the vacuum vector*.
- von Neumann algebra $\mathcal{A}(I) \subseteq \mathcal{B}(\mathcal{H}_0)$ for every closed interval $I \subset S^1$.

satisfying the following axioms:

- If $I_1 \subseteq I_2$, then $\mathcal{A}(I_1) \subseteq \mathcal{A}(I_2)$.
- **Covariance:** For any $\phi \in \text{Diff}^+(S^1)$, we have $u_\phi \mathcal{A}(I) u_\phi^* = \mathcal{A}(\phi(I))$ and if $\phi(p) = p$ for all $p \in I$, then $u_\phi x u_\phi^* = x$ for all $x \in \mathcal{A}(I)$.
- **Locality:** If $I_1 \cap I_2 = \emptyset$ then $[\mathcal{A}(I_1), \mathcal{A}(I_2)]_{\mathcal{B}(\mathcal{H}_0)} = 0$.
- u is a positive energy representation (the spectrum of L_0 is positive).
- Ω is cyclic for the algebra generated by all the $\mathcal{A}(I)$ inside $\mathcal{B}(\mathcal{H})$ and Ω is fixed under the action of the Möbius group.

Consequences of the axioms of conformal nets

Some important consequences are:

- **Haag Duality:** $\mathcal{A}(S^1 \setminus I) = \mathcal{A}(I)'$ in $\mathcal{B}(\mathcal{H}_0)$.
- **The Reeh-Schlieder theorem:** Ω is cyclic for every $\mathcal{A}(I)$ individually.
- We can make sense of $\mathcal{A}(I)$ for an abstract closed interval I as an abstract von Neumann algebra (not canonically embedded in $\mathcal{B}(\mathcal{H}_0)$), it assembles as a **covariant functor** $\mathcal{A}: \text{INT} \rightarrow \text{VNA}l\text{g}$ taking orientation-preserving embeddings to injective homomorphisms and orientation-reversing embeddings to injective antihomomorphisms.

- A part of the data of chiral CFT can be written as a modular functor, defined on the category of Riemann surfaces with marked points and holomorphic maps with sewing constraints.
- To give a functorial description of a chiral CFT, one needs to give a functor out of $\text{Cob}_{2,1}^{\text{conf}}$.

The following assignment ([Hen20]) is sketched inspired by the ideas in Graeme Segal's work [Seg04].

A linear category $\mathcal{C}(S)$ for every circle S .	A forgetful functor $U: \mathcal{C}(S) \rightarrow \text{Hilb}$
a functor $F_\Sigma: \mathcal{C}(S_{\text{in}}) \rightarrow \mathcal{C}(S_{\text{out}})$ for every cobordism $\Sigma: S_{\text{in}} \rightarrow S_{\text{out}}$	a bounded linear map $U_{\text{in}}(H) \rightarrow U_{\text{out}}(F_\Sigma(H))$ for every object $H \in \mathcal{C}(S_{\text{in}})$
trivialisations of F_Σ when Σ is an annulus	a certain holomorphicity condition

This data can be seen as data of an anomaly and a twisted or anomalous (conformal) quantum field theory.

Target for functorial chiral CFT

A chiral functorial CFT is a symmetric monoidal functor of 2-categories

$$Z: \text{Cob}_{2,1}^{\text{Conf}} \rightarrow W^*\text{-Cat}^{\text{concrete}}$$

- $W^*\text{-Cat}^{\text{concrete}}$ objects concrete W^* -categories, and morphisms concrete functors and natural isomorphisms.
- Thus, conformal automorphisms of complex cobordisms get assigned natural transformations.
- $W^*\text{-Cat}$ as a 2-category is equivalent to the Morita category of von Neumann algebras, pointed bimodules and bimodule homomorphisms, denoted $\text{vNAlg}^{\text{Mor}}$.
- These are models for 2-Hilb as they deloop the category Hilb.

- If W^* -category is a completely additive, admits a set of generators, then it is equivalent to the category of all (not necessarily separable) modules of some (not necessarily separable) von Neumann algebra, **R-Mod**.
- Every W^* -category $\mathcal{M}$ admits a canonical **Hilb-valued inner product**.
- If $\mathcal{M}$ is *nice* W^* -category, then $m \mapsto \langle -, m \rangle_{\text{Hilb}}$ defines an equivalence of categories between $\mathcal{M}$ and the category of completely additive functors $\overline{\mathcal{M}} \rightarrow \text{Hilb}$, categorified **Riesz representation theorem**.
- $\text{Endo}(\mathcal{M})$ is a **bi-involutive tensor category**, with involution $F \rightarrow \overline{F}$ determined by $\langle F(m_1), m_2 \rangle_{\text{Hilb}} \simeq \langle m_1, \overline{F}(m_2) \rangle_{\text{Hilb}}$.

Extending down to a point

- The fully extended conformal cobordism category $\text{Cob}_{2,1,0}^{\text{Conf}}$ is a delooping of $\text{Cob}_{2,1}^{\text{Conf}}$.
- The suitable target category would deloop $\text{vNAlg}^{\text{Mor}}$, we consider the morita category of **categorified von Neumann algebras** (first introduced in [Hen17]) called bicommutant categories.
- We denote this target of extended chiral CFTs as BicommCat .
- Since bicommutant categories are defined analogously to von Neumann algebras (as double commutant closed monoidal objects), this could be a candidate for 3-Hilb.

Outline of the assignment

Dim	χ -CFT
Point	Bicommutant Category
1-manifolds	Concrete Bimodule categories over Bicommutant Categories
Riemann surface surfaces	Concrete Bi-equivariant functors between bimodule categories
Conformal automorphisms of Riemann surfaces	Natural isomorphisms

Bicommutant Categories

For a bi-involutive tensor category $\mathcal{C}$

- An object Ω is **absorbing** if it is not a zero-divisor and for any non-zero divisor X , we have $\Omega \otimes X \simeq \Omega$. We denote the (idempotent completion of) the category of absorbing objects by $\text{Abs}(\mathcal{C})$.
- The tensor product gives us actions $L: \mathcal{C} \rightarrow \text{Endo}(\text{Abs}(\mathcal{C}))$ and $R: \mathcal{C}^{\text{op}} \rightarrow \text{Endo}(\text{Abs}(\mathcal{C}))$.
- We say a bi-involutive tensor category is **bicommutant** if it admits absorbing objects, its underlying dagger category is completely additive W^* -category with a set of generators and L, R are faithful bi-involutive tensor functors, and **the left and right actions are commutants of each other**.
- Compatibility of all structures is additional data given as unitary isomorphisms that satisfy some commutative diagrams.
- A canonical Hilb-valued **trace** $\text{Tr}: \mathcal{T}^{\text{abs}} \times \mathcal{T}^{\text{abs}} \rightarrow \text{Hilb}$ given by $\text{Tr}(A, B) = \langle \overline{A}, B \rangle_{\text{Hilb}}$ with *nice* properties like cyclicity and compatibility with the $\mathcal{T}$ action.

Modules over bicommutant categories

- Given a bicommutant category $\mathcal{C}$, a module over it is a W^* -category $\mathcal{M}$ equipped with a bi-involutive tensor functor $\mathcal{C} \rightarrow \text{Endo}(\mathcal{M})$.
- Given modules $\mathcal{M}_{\mathcal{C}}$ and ${}_{\mathcal{C}}\mathcal{N}$, we **Connes fuse** them as

$$\mathcal{M} \boxtimes_{\mathcal{C}} \mathcal{N} = \mathcal{M} \otimes_{\mathcal{C}} \text{Hom}_{\mathcal{C}\text{-Mod}}(\text{Abs}(\mathcal{C}), \mathcal{N})$$

- This gives us the composition of 1-morphisms (work in progress).

The category of Solitons

For an interval I , we define the categories $T_{\mathcal{A}}^+(I)$

- The *objects* are von Neumann algebra endomorphisms $\rho: \mathcal{A}(I) \rightarrow \mathcal{A}(I)$ such that there is a $J \subseteq I$ such that $\partial_+ I \in J$ and $\rho(x) = x$ for all $x \in \mathcal{A}(J)$.
- A *morphism* $\rho \rightarrow \sigma$ is an $a \in \mathcal{A}(I)$ such that $a\rho(x) = \sigma(x)a$ for all $x \in \mathcal{A}(I)$ and there is a $K \subseteq I$ such that $\partial_+ I \notin K$ and $a \in \mathcal{A}(K)$.

We define a monoidal structure on this as

- For objects ρ, ρ' , define $\rho \otimes \rho' = \rho' \circ \rho$.
- For morphisms $a_i: \rho_i \rightarrow \sigma_i$ ($i = 1, 2$) define $a_1 \otimes a_2 = \sigma_2(a_1)a_2$.

and we define a dagger structure by defining $a^\dagger = a^*$.

We call this category *left localised* endomorphisms.

The category of Solitons

Let S be a circle and p, q be two points on it and let us have orientation reversing diffeomorphism $\phi: S \rightarrow S$ which preserves p, q . Define $\mathcal{T}_{\mathcal{A}}(S, p, q)$ by

- *Objects*: Hilbert spaces with compatible actions of $\mathcal{A}(I)$ for each $I \subset S$ such that $p \notin \overset{\circ}{I}$
- *Morphisms*: Bounded linear maps that intertwine all the actions
- *Tensor product*: For two objects P_1, P_2 in $\mathcal{T}_{\mathcal{A}}(S, p, q)$, define $P_1 \boxtimes P_2$ to be

$$\mathrm{Hom}_{\mathcal{A}(S_-)\text{-Mod}}(H_0, P_1) \otimes_{\mathcal{A}(S_+)} P_2$$

where S_+ and S_- are the parts of S from p to q and q to p respectively.

The equivalence of category of Solitons

Let I be an interval and $p \in \dot{I}$. Let S be a circle with two marked points p, q and $\iota: I \rightarrow S$ be an embedding such that $\iota(p) = p$. Let I^+ be the part of I to the right of p and I^- be the part of I to the left of p . Then we have equivalences

$$T_{\mathcal{A}}^+(I^+) \simeq T_{\mathcal{A}}^-(I^-)^{\text{op}} \simeq T_{\mathcal{A}}(S, p, q)$$

- Given point-preserving embeddings $(I, p) \rightarrow (I', p')$, the induced map (given by extending the endomorphism to a larger domain by identity as it is already the identity at the boundary away from p), is an equivalence.
- The Drinfeld centre of $T_{\mathcal{A}}(S, p, q) \simeq \text{Rep}_S(\mathcal{A})$.

Absorbing objects and Bicommutant property

- Let $\phi: I \rightarrow I$ be a non-surjective embedding such that there is a $J \subseteq I$ such that $\partial_+ I \in J$ and $\phi(p) = p$ for $p \in J$.
- Then the object $\mathcal{A}(\phi) \in T_{\mathcal{A}}^+(I)$ is absorbing.
- Absorbing objects are important because they control half-braidings.

Further we can show that for any interval, there is a diagram

$$\begin{array}{ccc} & (\text{Abs}(T_{\mathcal{A}}^+(I)))' & \\ E_2 \nearrow & & \searrow E_3 \\ (T_{\mathcal{A}}^+(I))' & \xleftarrow{E_1} & T_{\mathcal{A}}^-(I) \end{array}$$

such that E_1 is equivalence of categories.

Thus, $T_{\mathcal{A}}(I, p)$ is bicommutant for any conformal net $\mathcal{A}$.

The assignment

- Given a **point** p and an interval I (collar), we assign it the category $\mathbf{T}_{\mathcal{A}}^+(\mathbf{p}, \mathbf{I}^+)$.
- Given an **interval** (a cobordism between its endpoints), we assign it $\mathcal{A}(\mathbf{I})\text{-Mod}$. This has actions of $T_{\mathcal{A}}^+(I)$, and thus a left action of $T_{\mathcal{A}}(\partial_- I) \simeq T_{\mathcal{A}}^+(I)$ and a right action of $T_{\mathcal{A}}(\partial_+ I) \simeq T_{\mathcal{A}}^-(I)^{\text{op}}$

To show that the assignment is functorial, we prove that for two interval I_1, I_2 with a common boundary p

$$\mathcal{A}(I_1)\text{-Mod} \boxtimes_{T_{\mathcal{A}}(p)} \mathcal{A}(I_2)\text{-Mod} \simeq \mathcal{A}(I_1 \cup_p I_2)\text{-Mod}$$

Let S be a circle with two marked points p, q . If we have functoriality, the circle should be assigned

$$\mathcal{A}(S_+)\text{-Mod} \boxtimes_{T_{\mathcal{A}}(p) \otimes T_{\mathcal{A}}(q)} \mathcal{A}(S_-)\text{-Mod}$$

We are currently working on finding this W^* -category for arbitrary conformal nets.

This is particularly interesting because the assumptions of finite-index (rationality) and strong-additivity have been dropped.



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